

Task Management System

Group 2 submission for CSE 504: Software Development Methodology. This project delivers a collaborative CRUD-based web application using Java, React, PostgreSQL, Docker, and GitHub Actions CI/CD.

1. Objective Mapping

Selected Topic
Group 2: Task Management System

Assignment Goal
Build a simple CRUD web application with Docker and CI/CD.

2. Technical Requirements Alignment

Java 21

Spring Boot 3

React + TypeScript

PostgreSQL

Docker

GitHub Actions

3. Functional Scope

Module	Implemented Features
Authentication	Register, login, logout, session-cookie authentication
Workspace Collaboration	Create workspace, join by invite code, list members
Task Management	Create, view, update, delete tasks with status, priority, assignee, due date
Board View	Kanban columns: TODO, IN_PROGRESS, DONE
Comments	Task discussion thread for collaboration
Filtering	Search, status, priority, assignee, overdue filtering

4. Architecture Overview



5. Data Model

Entity	Main Fields
User	id, name, email, passwordHash, createdAt
Workspace	id, name, description, inviteCode, createdAt
WorkspaceMember	workspaceId, userId, role, createdAt
Task	id, workspaceId, title, description, status, priority, dueDate, assigneeId, creatorId, createdAt, updatedAt
Comment	id, taskId, authorId, body, createdAt

6. API Summary

Auth

- POST /api/auth/register
- POST /api/auth/login
- POST /api/auth/logout
- GET /api/auth/me

Workspaces

- GET /api/workspaces
- POST /api/workspaces
- POST /api/workspaces/join
- GET /api/workspaces/{id}/members

Tasks

- GET /api/workspaces/{id}/tasks
- POST /api/workspaces/{id}/tasks
- GET /api/tasks/{id}
- PUT /api/tasks/{id}
- DELETE /api/tasks/{id}

Comments

- GET /api/tasks/{id}/comments
- POST /api/tasks/{id}/comments

7. Docker Requirement

The repository includes a multi-stage Dockerfile and a Docker Compose file for local execution.

- Dockerfile builds React, packages Spring Boot, and creates a runtime image
- docker-compose.yml starts the application container and PostgreSQL

8. CI/CD Requirement

- **CI:** install frontend dependencies, run frontend tests, build frontend, run backend tests
- **CD:** build Docker image on main and publish to Docker Hub

9. Verification Status

- Frontend tests passed
- Frontend production build passed
- Backend integration tests passed
- Backend jar packaging passed
- Docker Desktop requires a WSL update before the container stack can run on this machine

10. Submission Artifacts

Artifact	Status
GitHub Repository	Project is prepared for repository submission
Documentation PDF	Generated from this HTML document
Docker Image / Instructions	Dockerfile, Compose, and Docker Hub workflow included

Note: Docker runtime validation is blocked by a machine-level WSL update requirement. The application code, tests, packaged backend jar, and container configuration are already prepared.